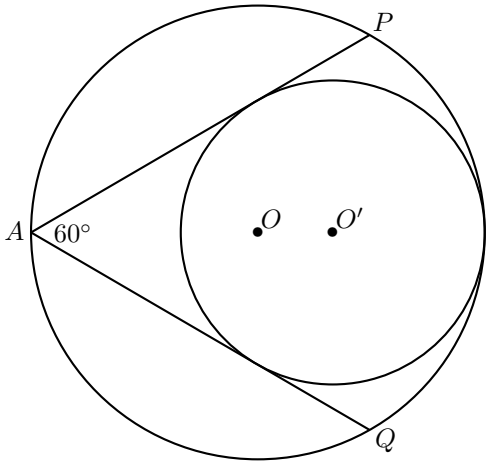
Trøndelagkonkurransen
Variety of Problems

1. In the cartesian plane, the graph of the parabola $y = x^2$ and the line $y = 2026$ bound a region S. How many points with integer coordinates are covered by S? (A point is considered covered by the region S if it lies on the boundary or inside S.)

2. Determine if there is a power of 2 such that, after rearranging its digits, it becomes another power of 2.

3. The figure below shows two circumferences of centers O and O' tangent to each other. A in the larger circumference is collinear with O and O'. The chords AP and AQ are tangent to the inner circumference. If $\angle QAP = 60°$, determine the ratio of the areas of the circles.



4. Find all real numbers a and b such that

$$a + b = 14,$$
$$a^3 + b^3 = 812.$$